

Faduma Osman

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EDUCATION

BS Biomedical Engineering

Joint Program at North Carolina State University & University of North Carolina at Chapel Hill

SKILLS

Solidworks, Project Management, Computer System Validation, Statistical Analysis, Lab Equipment Validation, Arduino, Benchling ELN, Matlab, Microsoft Excel, Microsoft Word, Microsoft Office Suite, Medical Device Development

EXPERIENCE

Operational Technology Intern

Kriya Therapeutics

Research Triangle Park, NC

May 2024-Dec 2024

- Conducted, reviewed, and verified Computer System Validation (CSV) deliverables including User Acceptance Testing (UAT), System Impact Assessment (SIA), Installation/Operational/Performance Qualifications (IQ/OQ/PQ), and User Requirements Specifications (URS) in compliance with 21 CFR Part 11 and GMP regulations
- Performed validation utilizing DOT compliance system to access and review Standard Operating Procedures (SOPs), ensuring adherence to regulatory guidelines during lab operations
- Customized and validated workflows within Benchling ELN platform, streamlining cross-functional team efficiency

Research Assistant

Gaddemhidi Circadian Clock and Genotoxic Stress Lab

Raleigh, NC

Sept 2021-Apr 2022

- Assisted in the design and execution of experiments, ensuring adherence to protocols and maintaining a high standard of laboratory safety
- Collected and processed biological samples, utilizing techniques such as DNA extraction, gel electrophoresis, and PCR

PROJECTS

Project Manager | IntraOcular Pressure Portable Air Puff Tonometer

Spring 2025

North Carolina State University

- Designed and developed a novel medical device to measure the pressure in the eye to detect glaucoma
- Documented design specifications with various validation deliverables to uphold FDA standards
- Utilized Solidworks and Arduino to create drawings for the CAD model

Project Manager | Adjustable Bone Plate Press

Fall 2024

North Carolina State University

- Used SolidWorks to design a novel biocompatible, sterilizable bone plate press capable of shaping patient-specific implants with precision.
- Developed a lever-driven mechanism and sliding plate array system to deliver controlled, high-force bending based on clinical use cases.
- Applied risk analysis and usability principles to support design documentation

Project Manager | Compressible Posterior End French-Eyed Needle

Fall 2023

North Carolina State University

- Designed a novel specialized surgical needle featuring a stainless steel body for durability and a posterior end made of flexible silicone to facilitate tissue compression
- Documented design specifications and testing results to demonstrate conformity with FDA regulations, facilitating regulatory approval and market readiness
- Utilized Solidworks to 3D model and render engineering drawings of prototype

Team Lead | Adhesive Bandage & Wound Sanitation Device

Spring 2023

North Carolina State University

- Developed a novel dual-compartment device for wound sanitation, following design controls and risk management standards
- Defined customer and product requirements per 21 CFR 820.30
- Managed document controls and design changes 21 CFR 820.40
- Utilized SolidWorks to create engineering drawings (ASME Y14.5) and 3D printing for prototypes